DSA0601 Data Handling and Visualization

Slot A

CO1 – Assessment Test 3 (AT3)

Visualization Design Exercise

Programming language: R (ggplot2)

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Question 1: Retail Data Visualization

1. Aim

To design one effective visualization of the supplied retail dataset that simultaneously represents Sales, Profit, Number of Customers, Product Category and Region, by choosing suitable aesthetic mappings and an appropriate scale transformation, and to justify every design decision.

2. Complete R Code

The code below is the content of Q1_Retail_Visualization.R and was executed to produce the figure(s) that follow.

```
software &lt;- data.frame(
Team=paste(&quot;T&quot;,1:20,sep=&quot;&quot;),
LinesOfCode=c(12000,18000,25000,35000,48000,
62000,81000,96000,120000,150000,
185000,240000,310000,390000,480000,
620000,810000,1050000,1350000,1800000),
Bugs=c(35,52,60,72,88,
110,125,142,165,188,
205,240,280,320,365,
420,490,560,640,720),
Developers=c(5,7,9,10,12,
14,16,18,20,22,
25,28,30,34,38,
42,46,50,55,60),
Complexity=c(2.1,2.4,2.7,3.0,3.2,
3.5,3.8,4.1,4.5,4.8,

5.1,5.4,5.8,6.2,6.6,
7.1,7.6,8.0,8.5,9.0)
)
```

3. Output Visualization

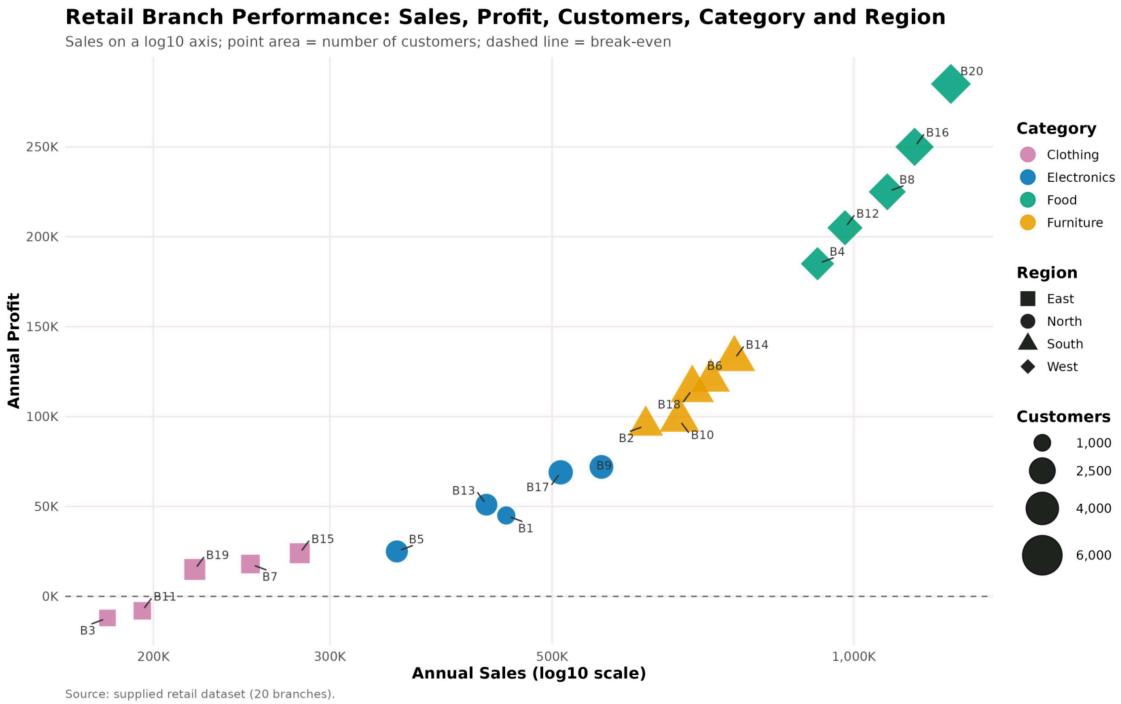


Figure 1.1: Retail branch performance. Sales on a log10 axis, profit on a linear axis, point area = customers, colour = category, shape = region (output of Q1_Retail_Visualization.R).

4. Interpretation of Results

The chart shows a very strong positive relationship between sales and profit. The correlation coefficient computed in R is 0.993, so branches with higher sales almost always earn proportionally higher profit. The point area, which encodes the number of customers, also grows steadily from lower left to upper right (correlation between sales and customers = 0.974), which indicates that the large-sales branches are also the branches that serve the most customers.

The four product categories form four visibly separate groups. **Food** branches (B4, B8, B12, B16, B20) occupy the upper right corner with sales between 920,000 and 1,250,000 and profits between 185,000 and 285,000. B20 is the strongest branch, with sales of 1,250,000, a profit of 285,000 and 6,100 customers. **Furniture** branches form a middle band, **Electronics** branches lie below them, and **Clothing** branches are clustered at the lower left with the smallest sales (180,000 to 280,000) and the smallest bubbles.

Only two branches lie below the dashed break-even line: B3 (profit of -12,000) and B11 (profit of -8,000). Both are Clothing branches in the East region. The average branch profit margin (Profit divided by Sales) is about 21.3% for Food, 16.1% for Furniture, 11.1% for Electronics and only 2.4% for Clothing, so category is the dominant factor behind profitability in this dataset.

At region level the totals are: West (Food) sales of 5,380,000 and profit of 1,150,000; South (Furniture) 3,460,000 and 560,000; North (Electronics) 2,300,000 and 262,000; East (Clothing) 1,125,000 and 37,000. It should be noted that in the supplied data every region sells exactly one category (North = Electronics, South = Furniture, East = Clothing, West = Food). Colour and shape are therefore perfectly redundant, and the data cannot separate a regional effect from a category effect.

5. Justification of Design Choices

- **Scatter plot (bubble chart).** Two quantitative variables with an expected relationship are best compared by position along common scales, which is the most accurately decoded visual channel (Cleveland and McGill, 1984). Sales and Profit were therefore given the x and y positions, and the three remaining variables were added as secondary aesthetics.
- **Log10 scale on Sales.** Sales span a range of about seven-fold (180,000 to 1,250,000) and the low-sales branches are numerous. On a linear axis the Clothing and Electronics branches would be crowded together on the left. A log scale gives equal distance to equal ratios, spreads the small branches and keeps the large ones in view.
- **Linear scale on Profit with a zero reference line.** Profit contains negative values (-12,000 and -8,000), and the logarithm of a negative number is undefined, so a log

transformation cannot be applied. A linear axis with a dashed break-even line keeps the losses visible and meaningful.

- **Point area for Customers.** Customers is a quantitative magnitude, so size is a suitable secondary channel. scale_size_area() makes the area (not the radius) proportional to the value and maps zero to zero area, which avoids exaggerating differences.
- **Colour for Category and shape for Region.** Both are nominal variables, so unordered channels are required. Colour is the strongest channel for the variable that explains most of the variation (Category). Shape carries Region. Branch labels were added with ggrepel so that individual branches can be identified without overplotting. Partial transparency (alpha = 0.85) reveals overlapping points.

6. Aesthetic Mappings and Scale Transformation

The mapping of each variable to a visual channel and scale is summarised below.

Variable	Aesthetic	Scale / transformation
Sales	x position	scale_x_log10(); labels in thousands (K)
Profit	y position	Linear continuous scale; dashed line at zero
Customers	Point area (size)	scale_size_area(); area proportional to customers
Category	Colour	Manual Okabe-Ito colour-blind-safe palette
Region	Point shape	Manual shapes: circle, triangle, square, diamond

The Okabe-Ito palette was chosen because its colours remain distinguishable for viewers with the common forms of colour-vision deficiency. Because shape duplicates the information already carried by colour in this dataset, the chart is also readable when printed in greyscale.

7. Conclusion

The visualization successfully encodes all five required variables in one chart. Profit increases with sales and customer numbers, Food branches in the West are the most successful, and Clothing branches in the East are the weakest, with the only two loss-making branches (B3 and B11). The log10 transformation of Sales and the area-scaled bubbles make the chart readable and fair, while the linear profit axis preserves the meaning of negative values.

Question 2: Energy Generation (Cartesian and Polar Coordinates)

1. Aim

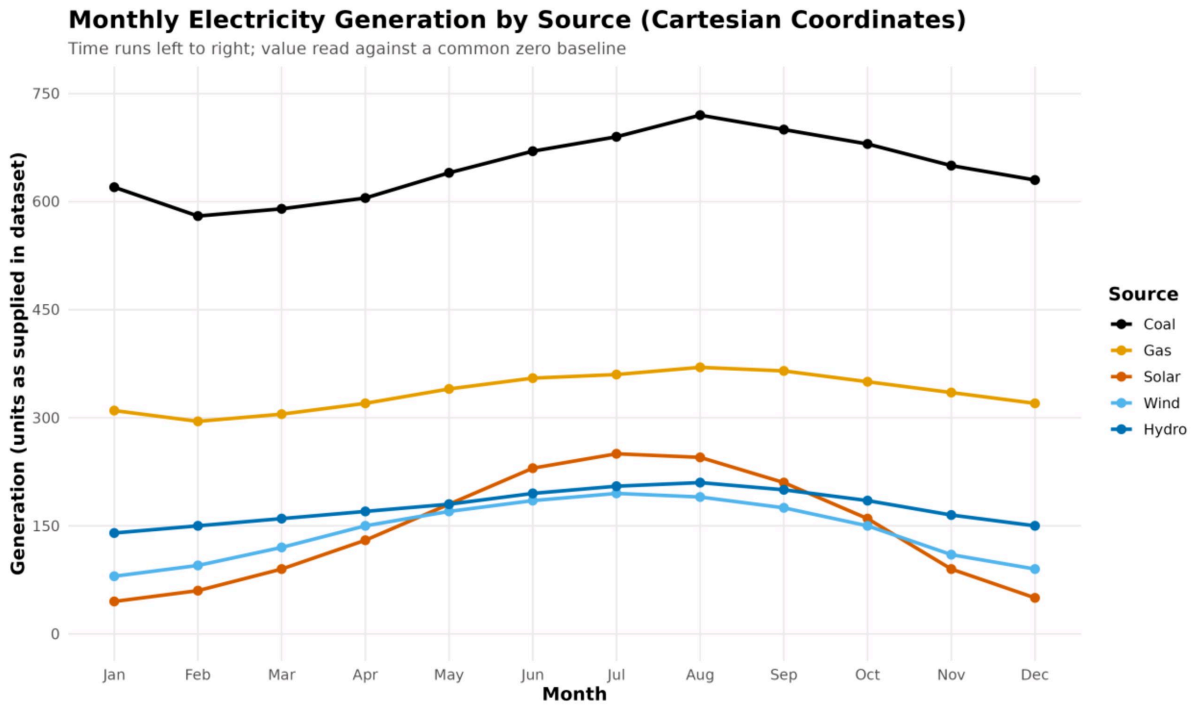
To visualize monthly electricity generation from five energy sources in both Cartesian and polar coordinate systems, to compare the two visualizations, and to determine which coordinate system provides better analytical insight.

2. Complete R Code

The code below is the content of Q2_Energy_Visualization.R and was executed to produce the figure(s) that follow.

```
energy &lt;- data.frame(
Month=month.abb,
Coal=c(620,580,590,605,640,670,690,720,700,680,650,630),
Gas=c(310,295,305,320,340,355,360,370,365,350,335,320),
Solar=c(45,60,90,130,180,230,250,245,210,160,90,50),
Wind=c(80,95,120,150,170,185,195,190,175,150,110,90),
Hydro=c(140,150,160,170,180,195,205,210,200,185,165,150)
)
```

3. Output Visualizations



kkkkkkkkkkkkkk

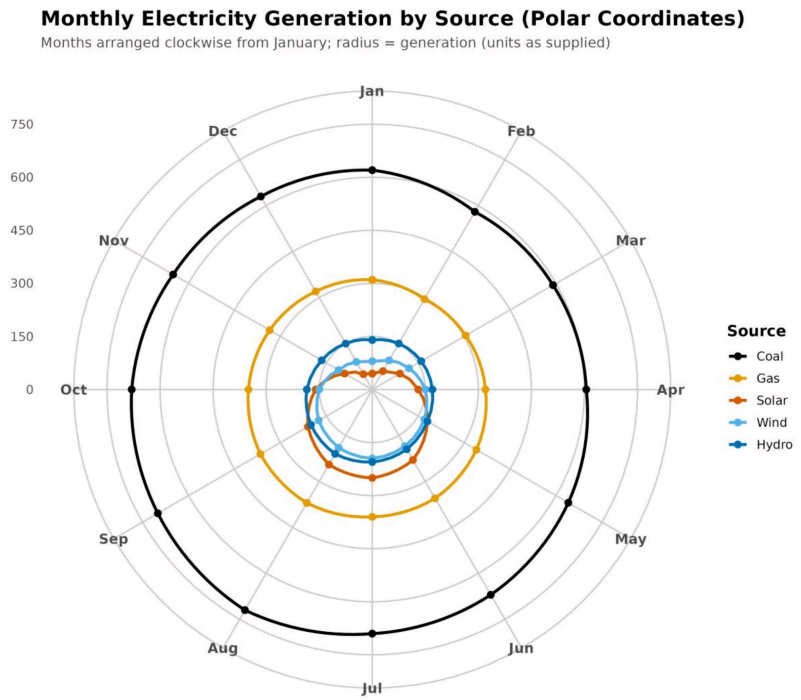


Figure 2.2: Monthly electricity generation by source in polar coordinates (radar-style line chart).

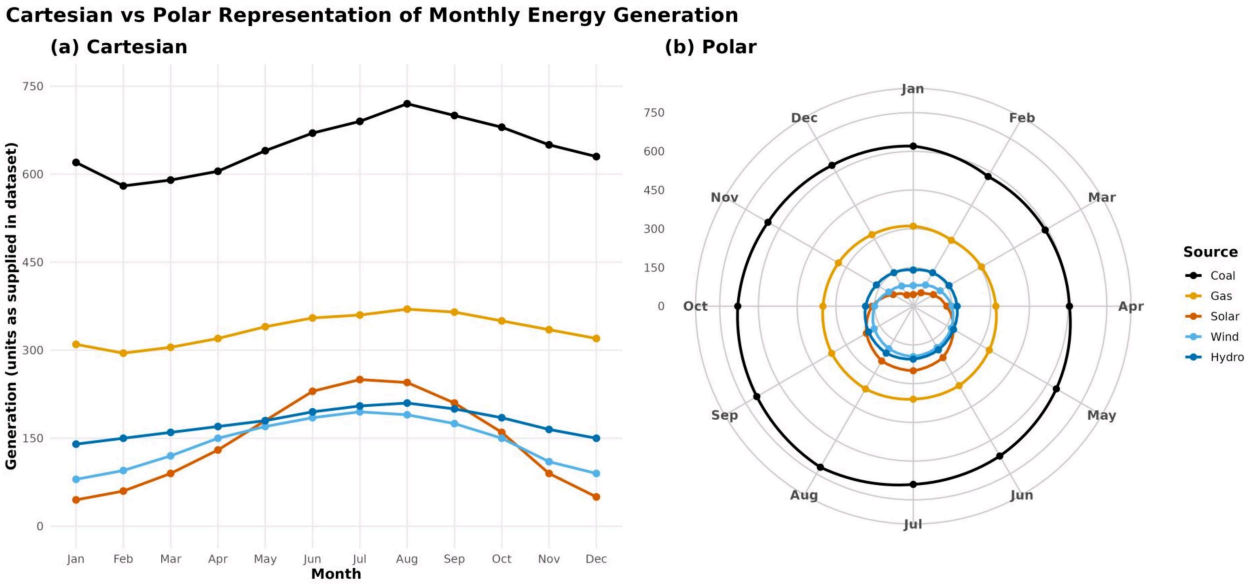


Figure 2.3: Side-by-side comparison of the Cartesian and polar representations.

4. Interpretation of Results

Coal is the dominant source in every month. It ranges from 580 in February to 720 in August and supplies 44.8% of the annual generation in the dataset. **Gas** follows the same seasonal shape at a lower level (295 in February to 370 in August) and supplies 23.2%. **Hydro** varies only moderately (140 in January to 210 in August, a ratio of 1.5) and supplies 12.2%.

Solar shows the strongest seasonality. It rises from 45 in January to 250 in July, which is a 5.56-fold change, and falls again to 50 in December. It supplies about 10.0% of the annual total. Solar overtakes both wind and hydro from June to September. **Wind** also peaks in July (195) and is lowest in January (80), a 2.44-fold change, and supplies 9.9%.

Total generation peaks in August (1,735) and is lowest in February (1,180). Every source reaches its maximum in July or August, so the summer months show the highest generation for the whole system.

In the Cartesian chart these facts can be read directly: the order of the sources is clear, the gaps between lines can be estimated against the y axis, and the points where solar crosses wind and hydro are easy to locate. In the polar chart the annual cycle is immediately visible as a closed loop for each source, with December joining smoothly to January. However, the three smallest sources (hydro, wind and solar) are compressed close to the centre, where their loops overlap, and it is difficult to read exact values or crossings.

5. Justification of Design Choices

- **Same encoding in both charts.** Both charts use lines with markers, the same y range (0 to 750), the same colours and the same legend, so that any difference in insight comes from the coordinate system alone and not from a change of chart type.
- **Zero baseline.** The radial and vertical axes both start at zero so that the lengths and distances are not distorted.
- **Closed polar loop.** January is repeated as a thirteenth point in the polar chart so that each series closes into a loop. Without this step a gap would appear between December and January.
- **Colour palette.** A colour-blind-safe palette was used: black for coal, orange for gas, vermilion for solar, sky blue for wind and blue for hydro. Coal is black because it is the dominant, conventional fossil source and needs the highest contrast.
- **Choice of line chart.** Line charts were used because the data form an ordered time series, and a line expresses continuity from month to month.

6. Coordinate Systems: Comparison

Cartesian coordinates map month to the horizontal position and generation to the vertical position. Polar coordinates map month to the angle (clockwise from January at the top) and generation to the distance from the centre. The two systems are compared in the table below.

Criterion	Cartesian	Polar
Accuracy of value reading	High: position along a common aligned scale	Lower: distance from a centre and angles are harder to judge
Comparing sources	Easy: parallel lines, crossings visible	Difficult: small sources overlap near the centre
Showing a yearly cycle	Adequate; the start and end are separated	Good: the loop returns to its starting point
Visual fairness	Neutral	Outer series (coal) appear exaggerated because the enclosed area grows with the square of the radius
Space efficiency	Uses a wide rectangle well	Needs a square area; large empty centre or corners
Overall suitability	Better for analysis	Better only for showing seasonality at a glance

Verdict. The Cartesian system provides better analytical insight for this dataset because the aim of the analysis is to compare magnitudes between sources and to identify peaks, troughs and crossings. The polar system is attractive and communicates the cyclical nature of the months, but this advantage is outweighed by the loss of precision and by the overlap of the smaller series. Polar coordinates are best used as a supplementary view for audiences who need an impression of seasonality.

7. Conclusion

Both coordinate systems display the same data, but they serve different purposes. The Cartesian line chart allows accurate quantitative comparison and shows that coal dominates, solar is the most seasonal source and total generation peaks in August. The polar chart emphasises the cyclical, seasonal pattern but compresses small values and makes precise comparison harder. Cartesian coordinates are therefore recommended as the primary analytical visualization, with the polar chart as a complementary illustration.

Question 3: Climate Data Visualization

1. Aim

To visualize Temperature, Air Quality Index (AQI), Rainfall and Humidity for 20 major Indian cities using appropriate colour scales, and to explain how the chosen palettes support accurate interpretation and accessibility.

2. Complete R Code

The code below is the content of Q3_Climate_Visualization.R and was executed to produce the figure(s) that follow.

```
climate &lt;- data.frame(
City=c(&quot;Delhi&quot;,&quot;Mumbai&quot;,&quot;Chennai&quot;,&quot;Kolkata&quot;,&quot;Hyderabad&quot;,
&quot;Bangalore&quot;,&quot;Pune&quot;,&quot;Ahmedabad&quot;,&quot;Jaipur&quot;,&quot;Lucknow&quot;,
&quot;Bhopal&quot;,&quot;Patna&quot;,&quot;Goa&quot;,&quot;Kochi&quot;,&quot;Nagpur&quot;,&quot;Indore&quot;,
&quot;Surat&quot;,&quot;Visakhapatnam&quot;,&quot;Coimbatore&quot;,&quot;Mysuru&quot;),
Temperature=c(44,35,38,40,39,31,33,43,45,39,
38,40,32,30,41,37,36,34,29,28),
Humidity=c(22,84,78,72,48,66,58,26,18,42,

36,45,89,91,30,34,76,82,68,64),
Rainfall=c(6,280,220,190,85,120,95,2,1,35,
48,72,360,410,12,20,180,240,130,145),
AQI=c(365,110,125,180,160,72,84,290,320,240,
175,220,58,52,270,190,145,98,60,55)
)
```

3. Output Visualizations

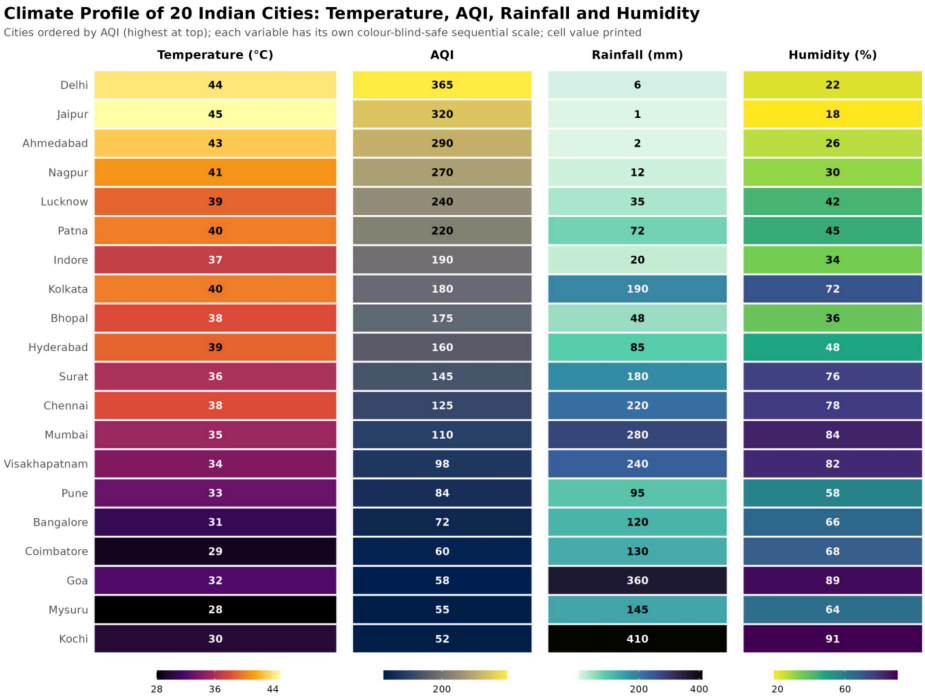


Figure 3.1: Multi-panel colour heat map of Temperature, AQI, Rainfall and Humidity for 20 Indian cities (cities ordered by AQI).

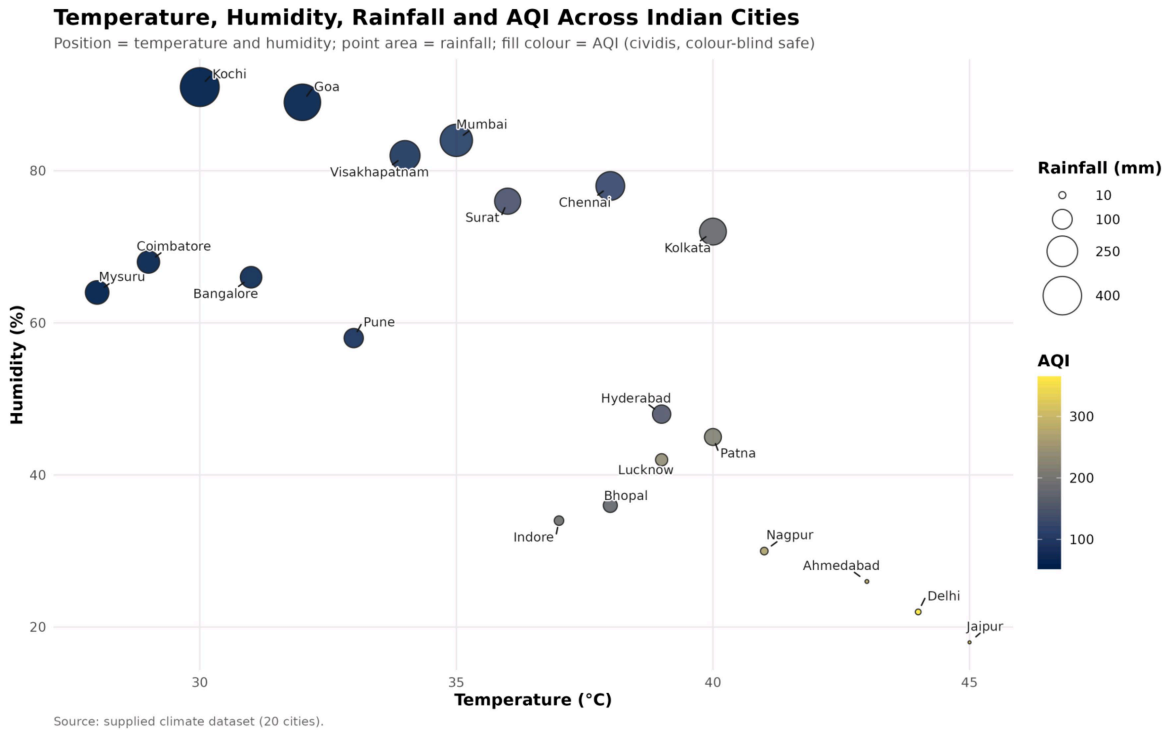


Figure 3.2: Temperature against humidity, with fill colour showing AQI and point area showing rainfall.

4. Interpretation of Results

The data range from 28 to 45 for temperature, 52 to 365 for AQI, 1 to 410 for rainfall and 18 to 91 for humidity. The heat map, with cities sorted from highest to lowest AQI, reveals two contrasting groups of cities.

Hot, dry and polluted cities. Delhi (AQI 365), Jaipur (320), Ahmedabad (290), Nagpur (270) and Lucknow (240) are at the top of the AQI ordering. These cities also have high temperatures (39 to 45), very low rainfall (1 to 35 for Delhi, Jaipur, Ahmedabad, Nagpur and Lucknow) and low humidity (18 to 42). Jaipur is the hottest city (45) and also the driest (rainfall of 1).

Cool, wet and clean cities. Kochi (AQI 52), Mysuru (55), Goa (58), Coimbatore (60) and Bangalore (72) have the lowest AQI values. They have lower temperatures (28 to 32), higher humidity (64 to 91) and, in the coastal cases, very high rainfall: Kochi has the highest rainfall (410) and the highest humidity (91), followed by Goa (rainfall 360, humidity 89).

The correlation matrix computed in R supports this reading. AQI is strongly positively correlated with temperature (0.94) and strongly negatively correlated with humidity (-0.85) and rainfall (-0.73). Humidity and rainfall are strongly positively correlated (0.93). These are associations among 20 cities and do not by themselves prove causation. The dataset also does not state units, so temperature is labelled in degrees Celsius and rainfall in millimetres as an assumption.

5. Justification of Design Choices

- **Heat map for a city-by-variable table.** The data form a matrix of 20 cities by 4 variables, and a heat map shows all 80 values together. Printing the value inside each cell lets readers obtain exact numbers, so colour supports pattern finding while the numbers support precision.
- **One scale per variable.** The four variables have different units and ranges, so a single shared colour scale would be meaningless. Each variable therefore has its own legend, and the variables are shown as separate panels with the same city order.
- **Sequential (not diverging) palettes.** Every variable is a quantity that increases from low to high without a natural midpoint, so sequential scales are appropriate. Diverging palettes would imply a meaningful central value that does not exist here.
- **Cities sorted by AQI.** Sorting by the variable of greatest public-health concern turns the heat map into an ordered display, and the associated variables then form visible gradients.
- **Second view.** The bubble chart (Figure 3.2) places temperature and humidity on the axes, encodes rainfall as area and AQI as colour, and labels each city. It complements the heat map by showing the inverse relationship between temperature and humidity and where the polluted cities fall.

6. Colour Scales, Accessibility and Interpretation

The palettes are taken from the viridis family, which was designed to be perceptually uniform, to be readable by viewers with colour-vision deficiency and to remain legible when printed in greyscale.

Variable	Palette	Direction and reason
Temperature	inferno	Dark (cool) to bright yellow (hot); a warm palette matches the intuitive meaning of heat
AQI	cividis	Dark blue (clean) to yellow (polluted); optimised for red-green colour-vision deficiency and avoids the traditional green-to-red AQI colours that are hard to separate for many viewers
Rainfall	mako (reversed)	Light (dry) to dark blue/black (very wet); blue tones are associated with water
Humidity	viridis (reversed)	Yellow (dry air) to deep purple (humid air); a distinct hue family so that this panel is not confused with the other three

Accessibility. About 1 in 12 men has some form of colour-vision deficiency, and rainbow or red-green scales are the usual cause of misreading. The chosen palettes vary monotonically in lightness, so the order of values can be recognised from brightness alone, even by colour-blind readers or in a black-and-white print. The numbers printed in the cells provide a redundant, non-colour encoding. The label colour (black or white) was selected automatically from the luminance of each cell to guarantee readable contrast.

Interpretation. Because lightness increases consistently with value in each panel, the reader can compare a city across the four panels and see the pattern immediately: the top rows combine bright temperature and AQI cells with pale rainfall cells and bright (dry) humidity cells, while the bottom rows show the reverse.

7. Conclusion

The multi-panel heat map, supported by the bubble chart, represents all four climate variables for the 20 cities using perceptually uniform, colour-blind-safe sequential palettes. It shows that hot, dry inland cities such as Delhi, Jaipur and Ahmedabad have the worst air quality, while humid, rainy and cooler coastal or southern cities such as Kochi, Goa and Mysuru have the best. The choice of one palette per variable, the printed cell values and the monotonic lightness make the display accurate and accessible.

Question 4: Software Development Data (Nonlinear Axis Transformation)

1. Aim

To visualize Lines of Code, Bugs, Developers and Complexity for twenty software teams using a nonlinear axis transformation suitable for highly skewed data, and to explain how the transformation improves interpretation compared with a linear scale.

2. Complete R Code

The code below is the content of Q4_Software_Visualization.R and was executed to produce the figure(s) that follow.

```
software &lt;- data.frame(
Team=paste(&quot;T&quot;,1:20,sep=&quot;&quot;),
LinesOfCode=c(12000,18000,25000,35000,48000,
62000,81000,96000,120000,150000,
185000,240000,310000,390000,480000,
620000,810000,1050000,1350000,1800000),
Bugs=c(35,52,60,72,88,
110,125,142,165,188,
205,240,280,320,365,
420,490,560,640,720),
Developers=c(5,7,9,10,12,
14,16,18,20,22,
25,28,30,34,38,
42,46,50,55,60),
Complexity=c(2.1,2.4,2.7,3.0,3.2,
3.5,3.8,4.1,4.5,4.8,

5.1,5.4,5.8,6.2,6.6,
7.1,7.6,8.0,8.5,9.0)
)
```

3. Output Visualizations

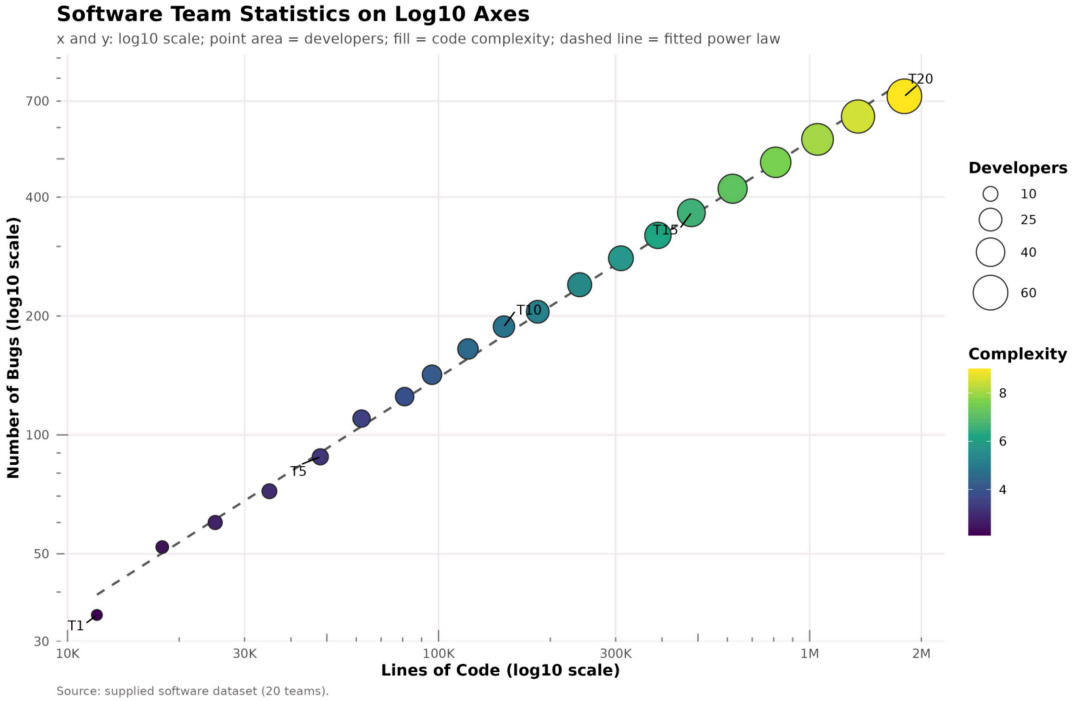


Figure 4.1: Bugs against Lines of Code on log10 axes (point area = developers, fill = complexity, dashed line = fitted power law).

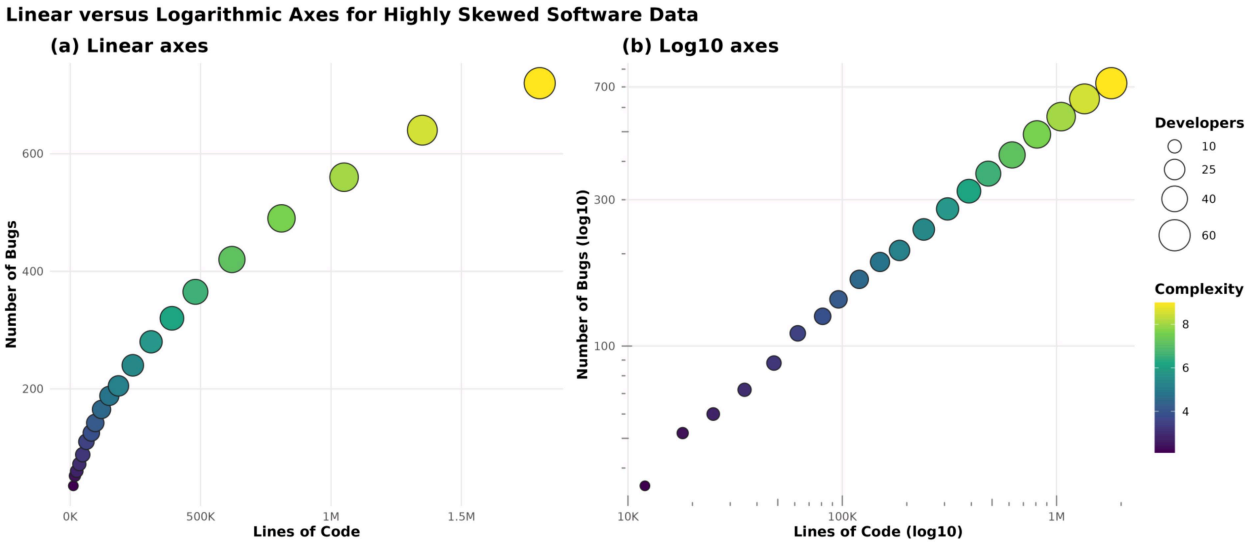


Figure 4.2: The same data on linear axes (a) and on log10 axes (b).

4. Interpretation of Results

The data are highly skewed. Lines of code range from 12,000 to 1,800,000, a 150-fold difference, and bugs range from 35 to 720, a 20.6-fold difference. Eleven of the twenty teams (55%) have fewer than 200,000 lines of code.

On the log10 chart (Figure 4.1) the twenty teams are spread evenly along a nearly straight line. A straight line on log-log axes indicates a power-law relationship. The regression of log10(Bugs) on log10(Lines of Code) calculated in R has a slope of 0.598 and an R-squared of 0.9974. This means that bugs increase roughly as the 0.6 power of code size: a ten-fold increase in code is accompanied by an approximately four-fold increase in bugs (10 raised to 0.598 is about 3.96). The relationship is sublinear, so the number of bugs per thousand lines of code falls from 2.92 for team T1 to 0.40 for team T20.

Colour and size show that complexity (2.1 to 9.0) and the number of developers (5 to 60) both increase steadily along the same line. Larger projects are therefore also more complex and have larger teams. Since these three variables rise together (Spearman correlation between lines of code and bugs is 1.0), this dataset cannot isolate the separate effect of each factor on bugs.

In the linear comparison (Figure 4.2a) the opposite problem is visible: the first eleven teams are squeezed into the leftmost 11% of the horizontal axis, with overlapping points, while the last few teams are far apart. On the log axis (Figure 4.2b) every team is clearly distinguishable.

5. Justification of Design Choices

- **Why a nonlinear transformation is required.** Both lines of code and bugs are right-skewed, covering more than two orders of magnitude for lines of code. On a linear axis the largest values dominate the scale and the small teams are compressed into a small region, so their pattern is lost.
- **Why the logarithm (base 10).** A log scale gives equal axis distance to equal ratios (for example 10,000 to 100,000 occupies the same distance as 100,000 to 1,000,000). This matches how software size is naturally discussed (orders of magnitude), converts multiplicative growth into additive growth, and turns a power-law relationship into a straight line that can be fitted and read easily. All values are positive, so the logarithm is defined for every team. A square-root transformation was considered but is too mild for a 150-fold range.
- **Log scale on both axes.** Both variables are skewed and the relationship is multiplicative, so both x and y use log10 scales, and annotation_logticks() marks the logarithmic intervals so that readers do not mistake the axes for linear ones.
- **Additional aesthetics.** Fill colour (viridis) shows Complexity because it is a continuous, ordered variable, and point area (scale_size_area) shows the number of developers. A

dashed linear fit (computed on the log scale) summarises the trend. Selected teams (T1, T5, T10, T15, T20) are labelled to give reference points without clutter.

6. Nonlinear Axis Transformation and Mappings

Variable	Aesthetic	Scale / transformation
Lines of Code	x position	scale_x_log10(); breaks at 10K, 30K, 100K, 300K, 1M, 2M
Bugs	y position	scale_y_log10(); breaks at 30, 50, 100, 200, 400, 700
Developers	Point area	scale_size_area()
Complexity	Fill colour	Continuous viridis scale (perceptually uniform)
Trend	Dashed line	Linear fit on the transformed (log-log) scale

Improvement over a linear scale. The fit of a straight line is better on the log-log scale (R-squared 0.9974) than on the linear scale (R-squared 0.9379), which shows that the multiplicative description is more faithful to the data. More importantly for readability, the log scale separates the crowded small teams and makes proportional differences comparable across the whole range.

7. Conclusion

A log10 transformation of both axes is the appropriate way to display this strongly skewed software data. It reveals a clean power-law relationship in which bugs grow approximately with the 0.6 power of code size, shows that larger projects also have more developers and higher complexity, and removes the crowding that makes the linear plot hard to read. The comparison in Figure 4.2 demonstrates the improvement in interpretability.

Question 5: Financial Data Visualization

1. Aim

To develop an integrated visualization that communicates Revenue, Profit Margin, Market Capitalization, Employee Strength and Industry Sector for twenty listed companies, using appropriate aesthetic mappings, coordinate systems, scales and colour palettes to maximise interpretability and minimise perceptual bias.

2. Complete R Code

The code below is the content of Q5_Financial_Visualization.R and was executed to produce the figure(s) that follow.

```
companies &lt;- data.frame(
Company=paste(&quot;C&quot;,1:20,sep=&quot;&quot;),
Sector=c(&quot;IT&quot;,&quot;Banking&quot;,&quot;Healthcare&quot;,&quot;Retail&quot;,&quot;Energy&quot;,
&quot;IT&quot;,&quot;Banking&quot;,&quot;Healthcare&quot;,&quot;Retail&quot;,&quot;Energy&quot;,
&quot;IT&quot;,&quot;Banking&quot;,&quot;Healthcare&quot;,&quot;Retail&quot;,&quot;Energy&quot;,
&quot;IT&quot;,&quot;Banking&quot;,&quot;Healthcare&quot;,&quot;Retail&quot;,&quot;Energy&quot;),
Revenue=c(420,580,720,850,980,
1150,1320,1480,1650,1820,
2050,2280,2550,2840,3150,
3480,3860,4250,4680,5200),
ProfitMargin=c(8.2,15.6,11.4,9.3,18.5,
20.4,17.8,14.2,10.5,22.1,
24.5,19.8,16.7,13.4,26.2,
28.1,21.5,18.3,15.2,30.4),
Employees=c(1500,3200,2100,4200,2800,
5100,6800,3900,7200,4500,
8100,9500,6200,10300,7200,
11800,13200,8600,14500,16000),

MarketCap=c(8500,12000,9800,7600,14500,
18200,21500,16800,13200,24500,
28500,32500,24800,19200,38200,
42500,46800,35600,29800,52000)
)
```

3. Output Visualization

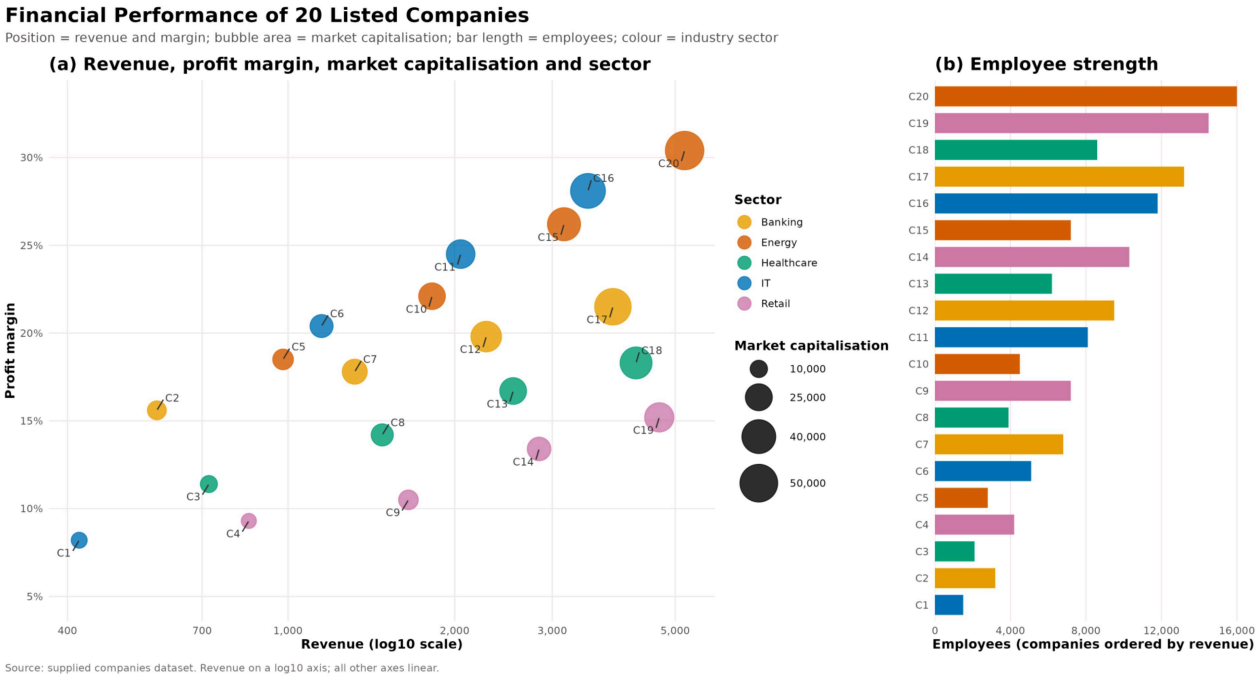


Figure 5.1: Integrated financial visualization: (a) bubble chart of revenue, profit margin, market capitalisation and sector; (b) employee strength by company.

4. Interpretation of Results

Revenue is positively related to the other measures. The correlation of revenue with market capitalisation is 0.877, with employees 0.914 and with profit margin 0.582. Larger companies therefore tend to be more valuable and to employ more people, and they also tend to be somewhat more profitable.

Company C20 (Energy) is the leading company on every measure: highest revenue (5,200), highest profit margin (30.4%), largest market capitalisation (52,000) and most employees (16,000). Company C1 (IT) has the lowest profit margin (8.2%) and the smallest revenue (420).

Sector differences are clear. **Energy** has the highest average profit margin (24.3%) and the highest average market capitalisation (32,300). **IT** has the lowest average revenue (1,775) but a high average margin (20.3%). **Banking** is in the middle (average margin 18.7%, average market capitalisation 28,200). **Healthcare** has an average margin of 15.2%. **Retail** has the lowest average margin (12.1%) and the lowest average market capitalisation (17,450) but the largest average workforce (9,050 employees), which suggests labour-intensive, low-margin businesses.

The bubbles in panel (a) rise from lower left to upper right, which agrees with the correlations above. Because the colours are the same in both panels, the reader can follow a sector from the bubble chart into the employee bars.

5. Justification of Design Choices

- **Bubble chart for four variables.** Position along x and y carries the two most important quantitative comparisons (revenue and profit margin), area carries market capitalisation, and colour carries the categorical sector.
- **Separate bar panel for employees.** A bubble chart that carried a fifth variable (for example through a second size or colour scale) would be overloaded. Bar length from a common zero baseline is the most accurate way to compare quantities (Cleveland and McGill, 1984), so employees are shown in a companion bar chart ordered by revenue rank so that the two panels can be read together.
- **Area-proportional bubbles.** scale_size_area() makes bubble area proportional to market capitalisation and maps zero to zero. Mapping the value to radius would exaggerate differences, which is a recognised source of perceptual bias.
- **Log10 revenue axis.** Revenue ranges from 420 to 5,200 (about 12-fold). The log scale spaces the small companies adequately and makes proportional differences comparable. Profit margin remains on a linear percentage axis because it is already a ratio.
- **Qualitative palette for sector.** Industry sector is nominal, so a categorical colour palette (Okabe-Ito, colour-blind safe) is used, with no implied order. The legend has enlarged keys so the sector colours remain identifiable on small bubbles.

- **Bias reduction.** Bars start at zero, there are no 3D effects or decorative gradients, the same colour is used for the same sector in both panels, and the labels (C1 to C20) allow exact identification of every company.

6. Aesthetic Mappings, Coordinates, Scales and Colours

Variable	Aesthetic / panel	Scale / transformation
Revenue	x position (panel a)	log10 scale, labelled breaks 400 to 5,000
Profit Margin	y position (panel a)	Linear scale, percentage labels
Market Capitalization	Bubble area (panel a)	scale_size_area(), area proportional to value
Industry Sector	Colour (both panels)	Manual Okabe-Ito qualitative palette
Employees	Bar length (panel b)	Linear scale from zero

Coordinate systems. Panel (a) uses ordinary Cartesian coordinates. Panel (b) uses coord_flip() so that the horizontal bars give room for readable company labels and the longest bars correspond to the largest employers. Polar coordinates were deliberately not used, because the comparison in Question 2 showed that they reduce the accuracy of quantitative reading.

7. Conclusion

The integrated figure presents all five required variables without overloading any single chart. Revenue and profit margin are shown by position, market capitalisation by area, sector by colour and employee strength by aligned bars. The visualization shows that revenue, market capitalisation and employment grow together, that Energy companies are the most profitable and valuable, and that Retail companies combine the lowest margins with the largest workforces. The use of a log revenue axis, area-proportional bubbles, zero-based bars and a colour-blind-safe palette keeps perceptual bias to a minimum.

Appendix A: Shared Helper Script (common_theme.R)

This helper file is loaded by all five scripts. It defines the common plot theme and the function that saves each plot as a 300 dpi PNG file.

```
# Shared theme and helper used by all five scripts
library(ggplot2)


theme_assignment <- function(base_size = 12) {
  theme_minimal(base_size = base_size) +
    theme(
      plot.title       = element_text(face = "bold", size = base_size + 4),
      plot.subtitle    = element_text(colour = "grey30", size = base_size - 1),
      plot.caption     = element_text(colour = "grey40", size = base_size - 3, hjust = 0),
      axis.title       = element_text(face = "bold"),
      legend.title     = element_text(face = "bold"),
      panel.grid.minor = element_blank(),
      plot.background  = element_rect(fill = "white", colour = NA)
    )
}


save_png <- function(plot, filename, width, height) {
  dir.create("images", showWarnings = FALSE)
  ggsave(file.path("images", filename), plot, width = width, height = height,
         dpi = 300, device = ragg::agg_png, bg = "white")
  cat("Saved:", file.path("images", filename), "\n")
}
```

References

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